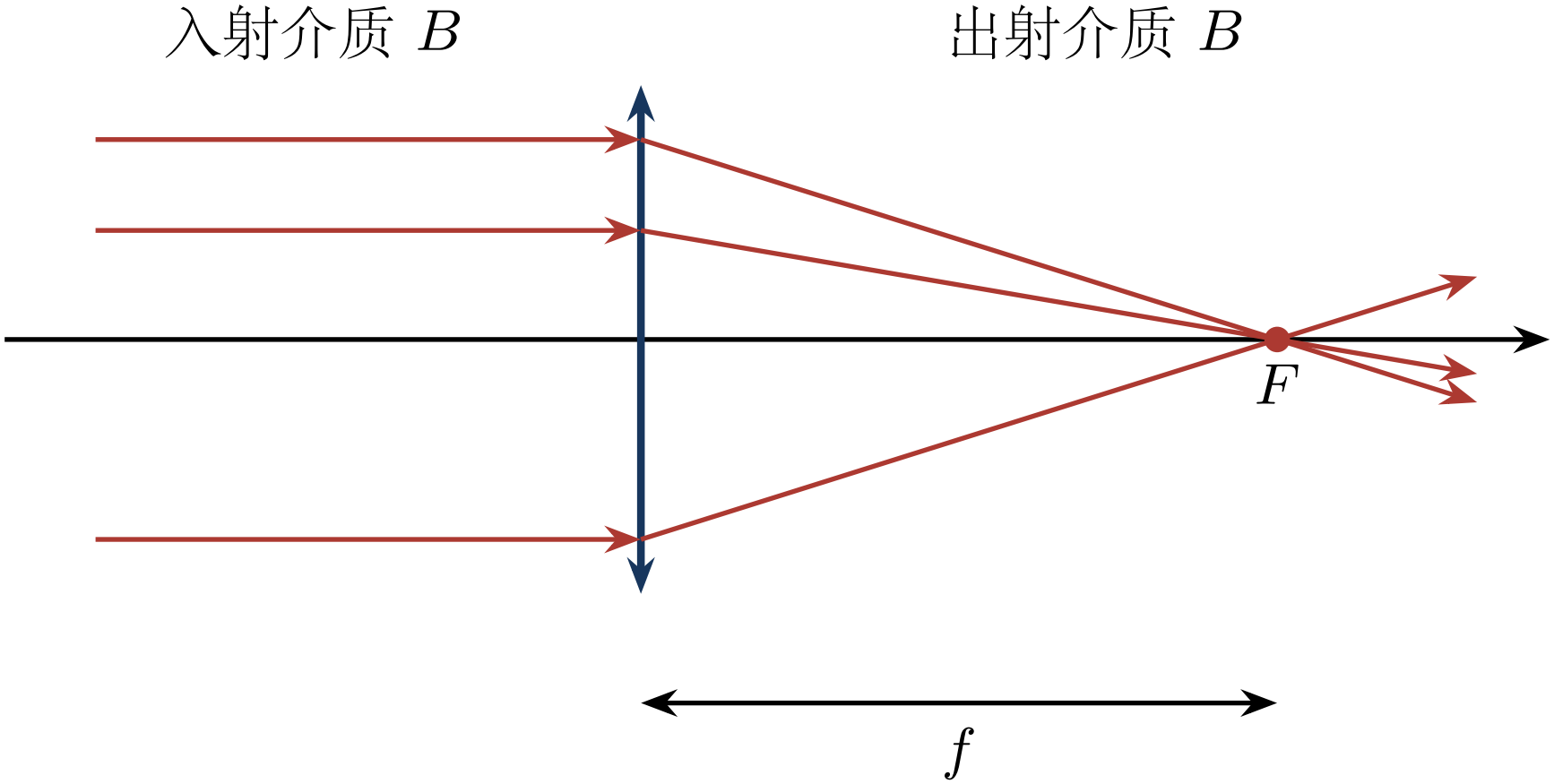

$$1/f \quad = \quad (n_A/n_B \; - \; 1)(1/R_L \; - \; 1/R_R)$$